



MEDICAL
UNIVERSITY

INNSBRUCK

Practical deep learning in medicine

Project intro

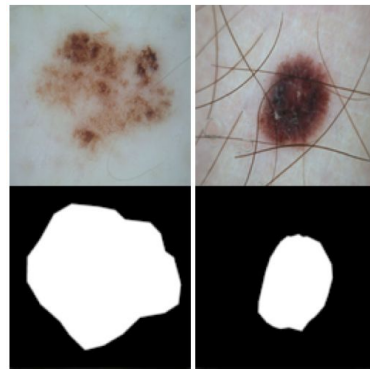
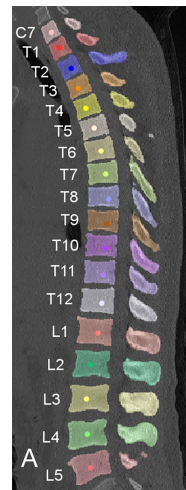
Session 5, 24.03.2026

Practical project

- Your opportunity to apply course knowledge to a novel project end-to-end
- All knowledge you need was covered in the course
- Project sessions
 - 7th, 14th, 21st, 28th April, 3-5pm, Computerraum C1
 - Come prepared with concrete questions
 - Continue to work on the project
- Work independently
 - Majority of project work happens outside of project sessions
- Presentation session
 - 5th May, 3-5pm, Computerraum C1

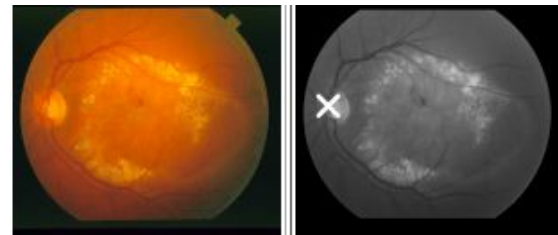
Suggested Projects - Segmentation

Project	Dimensions	Modality	Dataset link
Mitochondria segmentation	3D	Electron microscopy	MitoEM
Spine vertebrae segmentation	3D	CT	OSF
Covid lung segmentation	3D	CT	Zenodo
Cell instance segmentation	2D	Fluorescence microscopy	ISBI 2021
Skin lesion boundary segmentation	2D	Dermoscopy	ISIC 2018
Instrument segmentation (endoscopy)	2D	Endoscopy	Kvasir
Endoscopy multi-class segmentation	2D	Endoscopy	EndoCV
Gleason grading (pixel-level)	2D	Pathology	Gleason 2019



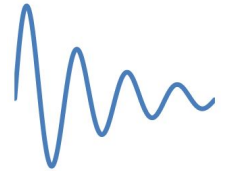
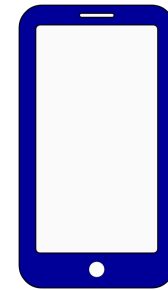
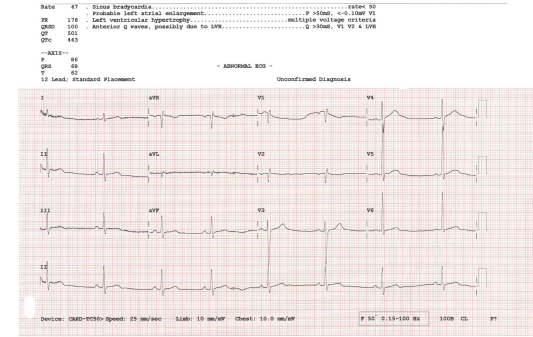
Suggested Projects - Imaging

Project	Task	Dimensions	Modality	Dataset link
Microaneurysm detection	Detection	2D	Retinal imaging	IDrID
Optic nerve detection	Detection	2D	Retinal imaging	STARE
Pneumonia detection	Detection	2D	X-ray	RSNA Kaggle
Age prediction from bone X-ray	Regression	2D	X-ray	RSNA Bone Age
Diabetic retinopathy grading	Classification	2D	Retinal imaging	Kaggle DR
Melanoma classification	Classification	2D	Dermoscopy	ISIC 2019
Gleason grading (core-level)	Classification	2D	Pathology	Gleason 2019



Suggested Projects - Signals

Project	Task	Dimensions	Modality	Dataset link
VO2 Max Prediction	Regression	1D	ECG	ECG in High Intensity Exercise Dataset
R-peak detection	Detection	1D	ECG	ECG in High Intensity Exercise Dataset
Atrial Fibrillation detection	Classification	1D	ECG	2017 PhysioNet Challenge
Apnea Detection	Segmentation/Classification	1D	ECG and Respiratory	Apnea-ECG Database
Heart Sound Classification	Classification	1D	PCG	2016 PhysioNet Challenge
Human Activity Recognition	Segmentation/Classification	1D	Accelerometer + Gyroscope	Human Activity Recognition using Smartphones



Suggested Projects - Own project

- Bring your own project
 - Be realistic about your goal
 - Talk to one of us beforehand
 - Make sure not to upload sensitive data to Google Colab!
 - Optional: Get inspired at <https://grand-challenge.org/challenges/> and look for datasets interesting to you!

Project Scope & Tips

- We want you to be able to complete something during the time you have
- If the datasets are too large to easily work on, use a subset! Or simplify the task - you can always add more complexity later]
- If you want, you can explore specific aspects of model performance
 - e.g., bias of AI algorithms for melanoma detection in different skin tones
- We suggest uploading the dataset to Google Drive and then linking the Colab to this data - Otherwise you'll need to upload the dataset every time!
- Don't just work on the project in the sessions! Come to each session with questions/areas you are stuck

Project presentation

Format

- 7 minutes of presentation
- 3 minutes of Q&A
- Bonus points for: Sticking to your time and asking a question during the Q&A of your colleagues

Baseline model

- Reasoning behind initial architecture
- Baseline performance

Final model

- Architecture, loss, hyperparams
- Evaluation

Dataset & motivation

- Why this dataset?
- Interesting patterns in the data
- Preprocessing decisions
- Train / val / test splits

Development process

- What went wrong?
- Iterations from baseline to final
- Key questions that came up

Discussion

- Hardest problem solved
- Biggest learning
- What would you do with more time
- What would you do differently

Project presentation - grading

- Delivery & clarity
 - Is the presentation engaging, well-structured and easy to follow?
- Visual design
 - Are the slides clean and visually effective?
 - We **do not** mean you need to make fancy slides with lots of animations. This is more related to whether the slide design aids or hinders the scientific communication
- Ambition/Scope
 - How challenging is the project?
 - Is there a clearly defined research question/goal?
- Documentation & reasoning
 - Are clear justifications given for key decisions?
- Subject-matter knowledge
 - Can you show a solid understanding of the underlying concepts?
- Q&A
 - Can you accurately answer questions about your work?

Grading - What it isn't

- We are not evaluating how good your final model is
- It is more important that you show a good understanding of the development process, regardless of the final result
- We understand some of you will likely use LLMs (e.g. ChatGPT) to help in the development of your model or presentation, but we expect you to be able to explain and understand what you have done

Questions?